Practice Set 1

1. A disk queue contains the following track numbers in the order they are requested:
   98, 183, 37, 122, 14, 124, 65, 67.
   The disk head starts at track 53. Calculate the total head movement for each of the following disk scheduling algorithms:
   a. FCFS (First-Come, First-Served)
   b. SSTF (Shortest Seek Time First)
   c. SCAN (Assume the disk goes towards the higher numbered tracks first)
   d. C-SCAN
   The disk has 200 tracks numbered 0 to 199.
2. Peterson's Solution ensures mutual exclusion. Describe how it satisfies the three requirements of a critical section:
   Mutual Exclusion, Progress and Two processes, P1 and P2, are synchronized using Peterson's Solution. Assume the following:
   flag[0] = true (Process P1 wants to enter the critical section).
   flag[1] = true (Process P2 also wants to enter).
   turn = 1.
   Which process will enter the critical section first? Show the step-by-step execution of the algorithm.
3. Given the reference string:
   7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, Number of frames: 3
   Simulate the following algorithms and calculate the number of page faults:
   FIFO (First-In-First-Out)
   LRU (Least Recently Used)
   Optimal Page Replacement
4. Explain the differences between short-term, medium-term, and long-term schedulers. How do they contribute to system performance?
5. Differentiate between monolithic kernels and microkernels. What are the advantages and disadvantages of each architecture?
6. Discuss the trade-offs between preemptive and non-preemptive scheduling. Can non-preemptive scheduling lead to starvation? Provide examples.
7. A system has three resource types with the following state: Available = (3, 3, 2) Allocation Matrix = [(0, 1, 0), (2, 0, 0), (3, 0, 2), (2, 1, 1), (0, 0, 2)] Maximum Matrix = [(7, 5, 3), (3, 2, 2), (9, 0, 2), (2, 2, 2), (4, 3, 3)]
   Apply the Banker's Algorithm to determine if the system is in a safe state.
8. Consider the set of 5 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

If the CPU scheduling policy is SJF non-preemptive, calculate the average waiting time and average turn around time.